Assignment- 1

COMPUTER NETWORKS- PCCST501

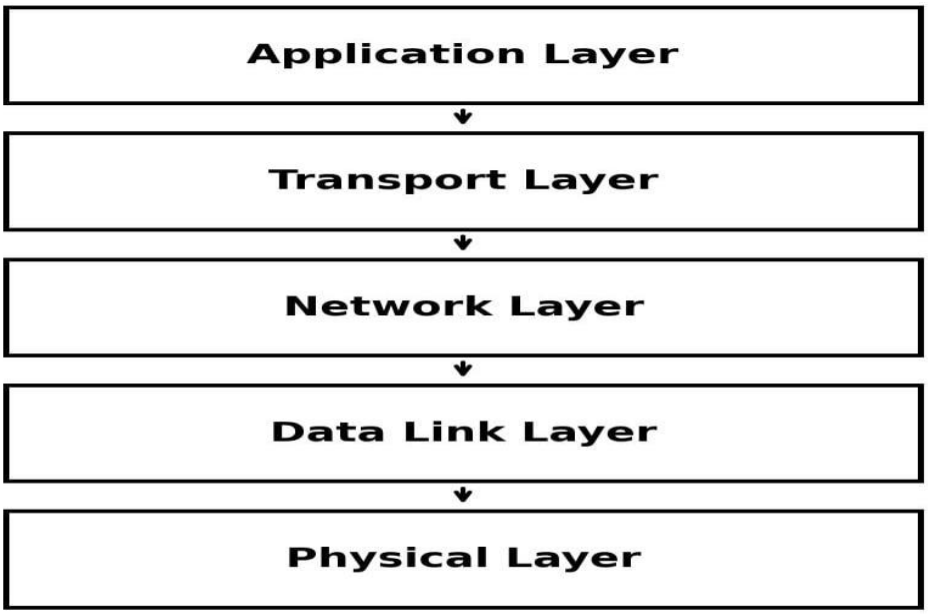
Submitted by,
Cristeena Boby
S5 CSA
Rollno.21

How the Internet Delivers a Web Page

Website Chosen: Amazon (https://www.amazon.com)

1. Draw the TCP/IP Protocol Stack and Explain the Role of Each Layer

The TCP/IP protocol suite is a five-layer architecture that enables communication between devices over the Internet. Each layer performs a specific function and provides services to the layer above it.



Application Layer

The application layer provides services directly to users and application programs. It enables communication between processes running on different hosts. Protocols such as HTTP, SMTP, FTP, DNS, and SSH operate at this layer. When a user opens Amazon's homepage, this is the layer that generates the HTTP request and interprets the returned HTML/CSS/JS.

Transport Layer

The transport layer provides process-to-process communication. It receives messages from the application layer, divides them into segments, and ensures reliable data delivery using TCP or fast communication using UDP. Amazon's web traffic uses TCP to guarantee that no part of the page is lost or arrives out of order.

Network Layer

The network layer is responsible for host-to-host communication. It uses the Internet Protocol (IP) to assign logical addresses and route packets through routers until they reach Amazon's data center.

Data Link Layer

The data-link layer transfers frames between adjacent devices on the same network. It adds MAC addresses and prepares data for transmission across the physical medium.

Physical Layer

The physical layer transmits bits as electrical, optical, or wireless signals over cables or air.

2. Explain the Role of DNS in Converting the Domain Name into an IP Address

Computers communicate using numeric IP addresses, but humans remember names like amazon.com far more easily. The Domain Name System (DNS) is the Internet's directory service that maps this human-readable name to Amazon's actual IP address. When the browser needs this mapping, it sends a query to a local/ISP DNS resolver. If that resolver does not already have the answer cached, it queries a chain of servers - a root server, a top-level-domain (.com) server, and finally Amazon's authoritative name server - which returns the correct IP address. This result is cached for a period defined by its Time-To-Live (TTL) value so that repeated visits to Amazon do not require a fresh lookup every time.

3. Describe How HTTP Is Used to Request and Receive the Webpage

HTTP (Hyper Text Transfer Protocol) is the standard protocol used to access web pages. It follows the client-server model, where the browser acts as the client and Amazon's web server acts as the server.

The communication takes place as follows:

- The browser sends an HTTP request to Amazon's web server.
- The request usually uses the GET method to retrieve the required webpage.
- The server processes the request.
- The server sends an HTTP response containing the requested webpage.

- The browser receives the response and displays the webpage.

4. Explain How TCP Provides Reliable Communication for HTTP

TCP provides:

- **Reliable delivery** :TCP guarantees that all data sent will reach the destination in the correct order.
- **Error control** : It detects missing or corrupted segments and resends them to ensure accuracy.
- **Flow control** : It regulates the sender's speed so the receiver can handle the data smoothly.
- **Congestion control** :It manages the transmission rate to prevent the network from becoming overloaded.

5. Identify Whether the Application Follows Client–Server or Peer-to-Peer Architecture and Justify Your Answer

- Amazon uses the **client–server model**.
- The browser acts as the **client**, Amazon's servers act as the **server**.
- Servers run continuously, waiting for requests; the client connects only when needed.
- The client sends an **HTTP request**, and the server processes it and replies with an **HTTP response**.
- Roles are fixed: the server must always be running, while the client disconnects after loading the page.
- Load is concentrated on Amazon's servers, unlike **peer-to-peer (P2P)** systems where users share the load.
- A P2P design wouldn't work for Amazon because catalogues, prices, inventory, and payments need **centralized, secure control.**

6. Compare HTTP and FTP

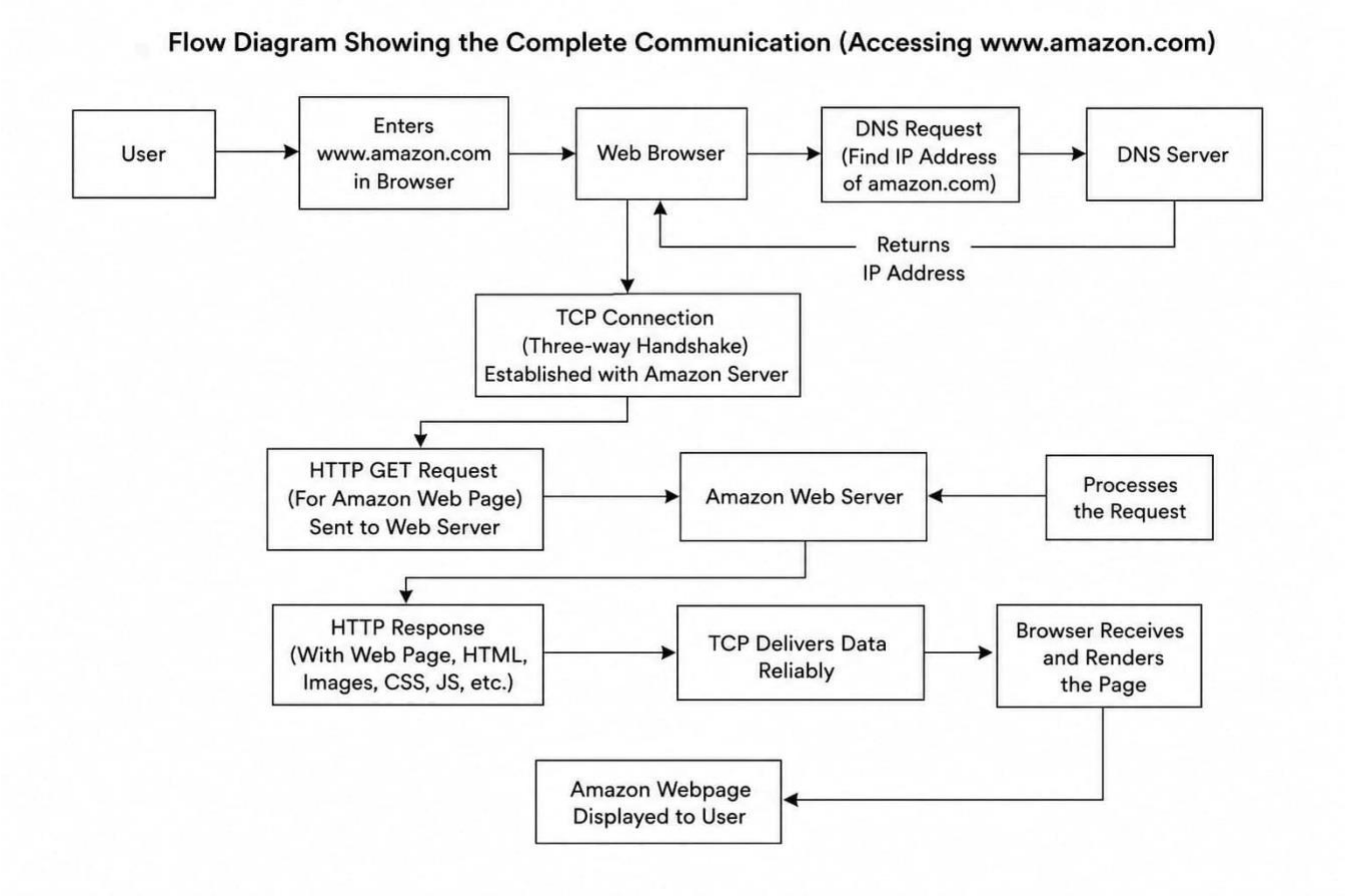
HTTP	FTP
Used to access web pages on the World Wide Web.	Used to transfer files between computers.
Uses a single TCP connection for communication (persistent or non-persistent).	Uses two TCP connections: one control connection and one data connection.
Mainly uses the GET and POST methods to request or send web resources.	Uses FTP commands such as RETR, STOR, LIST, and DELE for file operations.
Primarily transfers HTML pages, images, CSS, and JavaScript files.	Transfers files of different formats between local and remote systems.

7. Explain How Cookies Improve User Experience with One Real-World Example

Cookies improve user experience by remembering preferences, login details, and shopping cart items, so users don't have to re-enter information each time. They also help websites deliver personalized content and faster access, making browsing smoother and more convenient.

Real-world example: When a shopper adds a product to their Amazon cart, a cookie with a unique session ID is stored in the browser. Each time the shopper revisits or browses other pages, the cookie is sent back to Amazon's server, which reloads the cart and keeps the user logged in. This allows Amazon to provide a seamless shopping experience with saved carts and personalized recommendations.

8. Flow Diagram Showing the Complete Communication



9. Write a Brief Conclusion (100–150 words) Describing the Importance of Application-Layer Protocols in Everyday Internet Usage

Application-layer protocols are indispensable in everyday Internet usage because they provide the direct services users rely on, such as browsing, email, file transfer, and online shopping. Protocols like DNS, HTTP, and FTP simplify complex networking tasks by translating domain names into IP addresses, enabling communication between clients and servers, and supporting efficient data exchange. Mechanisms such as cookies further enhance usability by maintaining sessions and personalizing experiences, giving users a seamless interaction with websites. These protocols work in coordination with lower layers of the TCP/IP stack, which handle reliability, error control, and data delivery, ensuring that communication remains fast and consistent. Without standardized application-layer protocols, platforms like Amazon could not deliver secure transactions, consistent product catalogs, or personalized recommendations at global scale. In essence, application-layer protocols form the visible face of the Internet, bridging technical processes with user-friendly interactions that make modern online services possible.